

MTH2021 Mathematics Methods 2

Assignment 1 – solutions

January 23, 2022

1. If $z_1 = 3 - 2i$ and $z_2 = -1 + i$, then

(a)

$$\begin{aligned} z_1 - 2z_2 &= 3 - 2i - 2(-1 + i) \\ &= 5 - 4i \end{aligned}$$

(b)

$$\begin{aligned} z_1 z_2 &= (3 - 2i)(-1 + i) \\ &= -3 + 3i + 2i - 2i^2 \\ &= -1 + 5i \end{aligned}$$

(c)

$$\begin{aligned} z_1/z_2 &= \frac{3 - 2i}{-1 + i} \\ &= \left(\frac{3 - 2i}{-1 + i} \right) \left(\frac{-1 - i}{-1 - i} \right) \\ &= \frac{-3 - 3i + 2i + 2i^2}{1 - i^2} \\ &= -\frac{5}{2} - \frac{1}{2}i \end{aligned}$$

(d)

$$\begin{aligned} z_1 + z_2^* &= 3 - 2i + (-1 - i) \\ &= 2 - 3i \end{aligned}$$

2. Express the results of Question 1 in polar form $z = r \exp(i\theta)$.

(a) $5 - 4i$

$$\begin{aligned} r &= \sqrt{x^2 + y^2} \\ &= \sqrt{5^2 + 4^2} \\ &= \sqrt{41} \\ &= 6.403 \end{aligned}$$

$$\begin{aligned} \theta &= \text{Arctan}\left(\frac{y}{x}\right) \quad \text{since } x > 0 \\ &= \text{Arctan}\left(-\frac{4}{5}\right) \\ &= -0.6747 \end{aligned}$$

(b) $-1 + 5i$

$$\begin{aligned} r &= \sqrt{x^2 + y^2} \\ &= \sqrt{1^2 + 5^2} \\ &= \sqrt{26} \\ &= 5.099 \end{aligned}$$

$$\begin{aligned} \theta &= \pi + \operatorname{Arctan}\left(\frac{y}{x}\right) && \text{since } x < 0 \text{ and } y > 0 \\ &= \pi + \operatorname{Arctan}(-5) \\ &= 1.768 \end{aligned}$$

(c) $-\frac{5}{2} - \frac{1}{2}i$

$$\begin{aligned} r &= \sqrt{x^2 + y^2} \\ &= \frac{1}{2}\sqrt{5^2 + 1^2} \\ &= \frac{1}{2}\sqrt{26} \\ &= 2.550 \end{aligned}$$

$$\begin{aligned} \theta &= -\pi + \operatorname{Arctan}\left(\frac{y}{x}\right) && \text{since } x < 0 \text{ and } y < 0 \\ &= -\pi + \operatorname{Arctan}\left(\frac{1}{5}\right) \\ &= -2.944 \end{aligned}$$

(d) $2 - 3i$

$$\begin{aligned} r &= \sqrt{x^2 + y^2} \\ &= \sqrt{2^2 + 3^2} \\ &= \sqrt{13} \\ &= 3.606 \end{aligned}$$

$$\begin{aligned} \theta &= \operatorname{Arctan}\left(\frac{y}{x}\right) && \text{since } x > 0 \\ &= \operatorname{Arctan}\left(-\frac{3}{2}\right) \\ &= -0.9828 \end{aligned}$$

3. Sketch the following lines/curves/regions for z .

- (a) $z = 1 + it$ for $0 \leq t \leq 2$: a straight line joining the point $z = 1$ to the point $z = 1 + 2i$.
- (b) $z = t + it^2$ for $-1 < t < 1$: write $z = t + it^2 = x + iy$ giving the parabola $y = t^2 = x^2$ for $-1 < x < 1$.
- (c) $z = \cos t + i \sin t$ for $0 < t \leq 2\pi$: the circumference of the unit circle with centre at the origin.
- (d) $|z - 2i| = 2$: the circumference of a circle with radius 2 and centre at $z = 2i$.

(e) $|z + 1| \leq 1$: the closed region lying within and on the circumference of the circle with radius 1 and centre at $z = -1$.

4. If $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$, show that $(z_1 z_2)^* = z_1^* z_2^*$:

$$\begin{aligned}(z_1 z_2)^* &= ((x_1 + iy_1)(x_2 + iy_2))^* \\&= (x_1 x_2 - y_1 y_2 + iy_1 x_2 + ix_1 y_2)^* \\&= x_1 x_2 - y_1 y_2 - iy_1 x_2 - ix_1 y_2 \\&= (x_1 - iy_1)(x_2 - iy_2) \\&= z_1^* z_2^*\end{aligned}$$

5. We wish to prove the statement

$$(\cos \theta + i \sin \theta)^n = \cos(n\theta) + i \sin(n\theta)$$

It holds for $n = 1$. In an inductive proof, the second step is to show that if the result holds for $n = k$, then it holds for $n = k + 1$:

$$\begin{aligned}(\cos \theta + i \sin \theta)^{k+1} &= (\cos \theta + i \sin \theta)^k (\cos \theta + i \sin \theta) \\&= (\cos(k\theta) + i \sin(k\theta))(\cos \theta + i \sin \theta) \\&\quad \text{(result for } n = k\text{)} \\&= \cos((k+1)\theta) + i \sin((k+1)\theta)\end{aligned}$$

where for the last step we have used the relation

$$(\cos \alpha + i \sin \alpha)(\cos \beta + i \sin \beta) = \cos(\alpha + \beta) + i \sin(\alpha + \beta)$$

from lectures.

This establishes the result for $n = k + 1$. By mathematical induction, the result holds for all $n \geq 1$. It holds also for $n = 0$.

A quick way to prove it for negative integers $n = -k < 0$ is:

$$\begin{aligned}(\cos \theta + i \sin \theta)^{-k} &= \frac{1}{(\cos \theta + i \sin \theta)^k} \\&= \frac{1}{\cos(k\theta) + i \sin(k\theta)} \quad \text{(already proven)} \\&= \cos(k\theta) - i \sin(k\theta) \\&= \cos(-k\theta) + i \sin(-k\theta)\end{aligned}$$

Above we have used the relation

$$\begin{aligned}\frac{1}{\cos \phi + i \sin \phi} &= \frac{\cos \phi - i \sin \phi}{\cos^2 \phi + \sin^2 \phi} \\&= \cos(-\phi) + i \sin(-\phi)\end{aligned}$$

from lectures, which is just $1/\exp(i\phi) = \exp(-i\phi)$.

6. Use De Moivre's theorem to evaluate the expressions below.

(a) First

$$1 + i = \sqrt{2} \left(\frac{1}{\sqrt{2}} + i \frac{1}{\sqrt{2}} \right) = \sqrt{2} \exp(i\pi/4)$$

Then

$$(1 + i)^8 = 2^4 \exp(8\pi i/4) = 16$$

(b)

$$(1 - i) = (1 + i)^* \Rightarrow (1 + i)^8 - (1 - i)^8 = 16 - 16 = 0$$

(c) First

$$1 + i\sqrt{3} = 2 \left(\frac{1}{2} + i \frac{\sqrt{3}}{2} \right) = 2 \exp(i\pi/3)$$

Then

$$\begin{aligned} \frac{1 + i}{1 + i\sqrt{3}} &= \frac{\sqrt{2} \exp(i\pi/4)}{2 \exp(i\pi/3)} \\ &= \frac{1}{\sqrt{2}} \exp i(\pi/4 - \pi/3) \\ &= \frac{1}{\sqrt{2}} \exp i(-\pi/12) \\ &\Rightarrow \left(\frac{1 + i}{1 + i\sqrt{3}} \right)^{12} = \frac{1}{2^6} \exp i(-12\pi/12) = -\frac{1}{64} \end{aligned}$$

7. Use De Moivre's theorem to show that

$$\sin 4\theta = 4 \cos^3 \theta \sin \theta - 4 \cos \theta \sin^3 \theta$$

We need the binomial expansion for $(a + b)^4$:

$$(a + b)^4 = a^4 + 4a^3b + 6a^2b^2 + 4ab^3 + b^4$$

Then

$$\begin{aligned} (\cos \theta + i \sin \theta)^4 &= \cos^4 \theta + 4i \cos^3 \theta \sin \theta - 6 \sin^2 \theta \cos^2 \theta - 4i \cos \theta \sin^3 \theta + \sin^4 \theta \\ &= \cos 4\theta + i \sin 4\theta \end{aligned}$$

Equating real and imaginary parts:

$$\begin{aligned} \cos 4\theta &= \cos^4 \theta - 6 \sin^2 \theta \cos^2 \theta + \sin^4 \theta \\ \sin 4\theta &= 4 \cos^3 \theta \sin \theta - 4 \cos \theta \sin^3 \theta \end{aligned}$$

8. Find all distinct solutions of the following equations:

(a) $2z^2 + 2z + 1 = 0$

Using the formula for the solution of $az^2 + bz + c = 0$

$$z = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

gives

$$\begin{aligned} z &= \frac{-2 \pm \sqrt{4 - 8}}{4} \\ &= -\frac{1}{2} \pm \frac{1}{4}\sqrt{-4} \\ &= -\frac{1}{2} \pm \frac{1}{2}i \end{aligned}$$

(b) $z^4 = -1$

Write the RHS in polar form:

$$z^4 = -1 = \exp(i\pi)$$

The four distinct solutions for z are

$$z_k = \exp(i(\pi + 2k\pi)/4) \quad , \quad k = 0, 1, 2, 3$$

or

$$\begin{aligned} z_0 &= \frac{1}{\sqrt{2}}(1 + i) & z_1 &= \frac{1}{\sqrt{2}}(-1 + i) \\ z_2 &= \frac{1}{\sqrt{2}}(-1 - i) & z_3 &= \frac{1}{\sqrt{2}}(1 - i) \end{aligned}$$

(c)

$$z^3 = \sqrt{2} \frac{1+i}{\sqrt{2}} = \sqrt{2} \exp(i\pi/4)$$

$$\Rightarrow z_k = 2^{1/6} \exp(i\theta_k) \quad , \quad \theta_k = \frac{\pi/4 + 2k\pi}{3} \quad , \quad k = 0, 1, 2$$

giving the distinct solutions

$$\begin{aligned} z_0 &= 2^{1/6} \exp(i\pi/12) \\ z_1 &= 2^{1/6} \exp(3i\pi/4) \\ z_2 &= 2^{1/6} \exp(17i\pi/12) \end{aligned}$$

(d)

$$z^3 = -8 = 2^3 \exp(i\pi)$$

The three distinct solutions for z are

$$z_k = 2 \exp(i(\pi + 2k\pi)/3) \quad , \quad k = 0, 1, 2$$

or

$$\begin{aligned} z_0 &= 2 \exp(i\pi/3) = 1 + i\sqrt{3} \\ z_1 &= 2 \exp(i\pi) = -2 \\ z_2 &= 2 \exp(5i\pi/3) = 2 \exp(-i\pi/3) = z_0^* = 1 - i\sqrt{3} \end{aligned}$$